

Editorial

Urban Expansion Prediction and Land Use/Land Cover Change Modeling for Sustainable Urban Development

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1. Introduction

Urban expansion, a defining feature of the contemporary era, presents both challenges and opportunities for sustainable development [1]. This Editorial embarks on a comprehensive journey through recent developments in the field of urban expansion, revealing gaps in our understanding and showcasing how this Special Issue serves as a crucial milestone in bridging those knowledge gaps. Moreover, it will continue with a call to embrace the future by integrating artificial intelligence (AI) into urban expansion research, offering a transformative lens for understanding and navigating the complexities of urban growth.

Recent Developments in Urban Expansion: The past few decades have witnessed unparalleled urban growth, driven by factors such as population explosion, migration trends, and economic development [2]. Cities worldwide have experienced transformative changes in their spatial landscapes, raising pertinent questions about the sustainability and resilience of urban development [3]. Researchers and urban planners have grappled with the multifaceted nature of this expansion, leading to a surge in studies examining land use, ecological impact, and socio-economic ramifications [4,5]. While traditional approaches have shed light on many aspects of urban expansion, recent developments emphasize the need for more elegant methodologies. The integration of advanced technologies, such as machine learning, remote sensing, and spatial modeling, has opened new avenues for understanding the intricate dynamics of urban growth [6–8]. This Special Issue, comprising twelve articles, reflects this evolving landscape, offering a diverse set of insights that collectively contribute to a more comprehensive understanding of urban expansion.

Identifying the Gap in Knowledge: Despite significant progress, a notable knowledge gap persists, particularly in the holistic evaluation of urban expansion. Conventional metrics often fall short of capturing the complex interplay between ecological sustainability, socioeconomic considerations, and the ever-evolving urban landscape [8]. The need for a comprehensive framework that transcends traditional boundaries has become evident, urging researchers to explore innovative methodologies and interdisciplinary approaches. The gap extends to the understanding of urban types, change detection precision, and the dynamic nature of urban growth in diverse global contexts [9]. Moreover, the impact of urbanization on climate, green spaces, and vulnerable populations requires deeper exploration [10,11]. Recognizing these gaps is the first step toward informed and targeted research that comprehensively addresses the complexities of urban expansion.

How this Special Issue Addresses the Gaps: This Special Issue stands as a testament to the collective effort to address the identified gaps in urban expansion research. The twelve articles traverse a spectrum of topics, each offering a unique perspective and contributing to a more delicate understanding of the multifaceted challenges posed by urban growth.

Future Research: Incorporating Artificial Intelligence into Urban Expansion: As we find ourselves at the brink of a profound urban metamorphosis, the incorporation of artificial intelligence (AI) stands out as a central and compelling theme for prospective



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investigations in the realm of urban expansion. The advent of AI, endowed with its remarkable capacity to meticulously analyze extensive datasets, discern intricate patterns, and forecast outcomes, signifies a transformative prospect poised to redefine our comprehension and strategic approaches to the complex dynamics of urban development.

2. An Overview of Published Articles

2.1. *Reimagining Urban Growth: A Comprehensive Framework for Evaluating Land Use Efficiency in Tehran (1986–2021)*

This paper critically assesses the limitations of the land use efficiency evaluation formula proposed in the SDG 11.3.1 Indicator, with a focus on Tehran's urban expansion between 1986 and 2021. Contrary to the conventional understanding derived from land consumption and population growth, our research reveals that Tehran's urban landscape has not only diminished most urban services per capita, but has also compromised vital ecosystem services during this period. To address this dual challenge of sustainable urban development—ensuring robust urban services while safeguarding natural resources—a novel assessment framework is presented.

In this comprehensive framework, ten key variables, spanning environmental, physical–spatial, and economic–social domains, were identified as crucial contributors to the effects of urban expansion. Through collaboration with 14 urban planning experts, a questionnaire was formulated to capture the intricate relationships among these variables. Employing DEMATEL and interpretive structural modeling (ISM) methods, the data obtained from the questionnaires were analyzed to discern the extent to which various factors influenced and were influenced by urban expansion.

The findings unearth three distinct levels of influence within the framework. At the forefront of urban expansion are transportation, infill development, and entrepreneurship, identified as direct influencers. Spatial justice and housing/population attraction occupy an intermediate level, mutually shaping and being shaped by urban expansion. Finally, first-level variables such as land surface temperature, air pollution, sewage and waste, water resources, and vegetation emerge as primarily affected by the urban expansion dynamics.

By unraveling the intricate web of relationships among these variables, our proposed framework not only challenges the prevailing assumptions about Tehran's urban growth, but also provides urban planners with a delicate tool to enhance land use efficiency. This tool goes beyond the conventional understanding of urban expansion, emphasizing the importance of sustainable urban services and the responsible utilization of ecosystem services for a resilient and cost-beneficial urbanization process.

2.2. *Unveiling Urban Types: A Machine Learning Approach to Urban Zone Classification in Nonthaburi, Thailand, Using Integrated SAR and Optical Images*

The inexorable urbanization and expansion in emerging countries underscore the need for a delicate understanding of urban landscapes. This study, situated in Nonthaburi, Thailand, leverages machine learning methods to classify urban zones based on building height, estimated through the integration of Sentinel-1 synthetic aperture radar (SAR) and various satellite-based indices from Sentinel-2A.

The first objective involves estimating building height using a novel indicator, vertical–vertical–horizontal polarization (VVH), derived from dual-polarization information (VV and VH) of Sentinel-1 SAR. The resultant building height model, with a root mean square error (RMSE) of 1.413 m, lays the foundation for the subsequent urban classification. The second objective categorizes urban zones into three classes: residential buildings, commercial buildings, and other structures encompassing vegetation, water bodies, and car parks. Spectral indices including the normalized difference vegetation index (NDVI), normalized difference water index (NDWI), and normalized difference built-up index (NDBI) are extracted from Sentinel-2A data.

Three machine learning classifiers—support vector machine (SVM), random forest (RF), and k-nearest neighbor (KNN)—are employed for classification, using randomly trained data from a 500 m focus study subdivided into a 100 × 100 m grid. Different

models are explored, including those using only building height, only spectral indices, and a combination of both. Sixteen variables, comprising minimum, maximum, mean, and standard deviation from the NDVI, NDWI, NDBI, and building height, are utilized. Principal components analysis (PCA) is employed to enhance model performance.

The results demonstrate the efficacy of SVM, outperforming RF and KNN with accuracies of 0.86, 0.75, and 0.76, respectively. The integration of SAR-derived building height and optical image spectral indices not only refines urban classification, but also underscores the potential of machine learning in advancing our ability to discern and categorize complex urban landscapes. This approach contributes valuable insights for future land evaluations and urban planning in the face of rapid urbanization.

2.3. Unveiling the Geological Tapestry: Integrated Remote Sensing for Mapping Geological Units in the Saka Region, Northeast Morocco

In tandem with traditional geological survey data, satellite imagery proves to be a valuable asset for comprehensive geological mapping. This study focuses on the Saka region in the northeast part of Morocco, utilizing Landsat Oli-8 and ASTER images to map geological units. The primary objectives include (1) mapping the lithological facies of the Saka volcanic zone, (2) discriminating different minerals using Landsat Oli-8 and ASTER imagery, and (3) validating the results through field observations and geological maps.

The methodology encompasses a suite of techniques, including color composition (CC), band ratio (BR), minimum noise fraction (MNF), principal component analysis (PCA), and spectral angle mapper (SAM) classification. The integration of these techniques facilitates the achievement of the study's objectives. The results exhibit a robust discrimination between various lithological facies, corroborated by the supervised classification of images and validated through field missions and geological maps at a 1/500,000 scale.

The classification outcomes reveal the dominance of Basaltic rocks in the study area, followed by Trachy andesites and Hawaites. Surrounding these geological formations are quaternary sedimentary rocks, accompanied by an abundance of Quartz, Feldspar, Pyroxene, and Amphibole minerals. The successful integration of remote sensing data, supported by ground truthing and geological mapping, not only enhances our understanding of the geological composition of the Saka region, but also showcases the efficacy of satellite imagery in geological mapping endeavors. This integrated approach holds promise for advancing geological studies and resource assessments in similar regions globally.

2.4. Unveiling Change in Spectral Dimensions: A Multi-Dimensional Deep Siamese Network for Bi-Temporal Hyperspectral Imagery Change Detection

This study introduces an innovative change detection (CD) framework designed for bi-temporal hyperspectral imagery, leveraging a multi-dimensional deep Siamese network. The proposed method comprises two key steps: (1) the automated generation of training samples through the Otsu algorithm and the dynamic time wrapping (DTW) predictor, and (2) binary CD using a multidimensional convolution neural network (CNN). The evaluation of this approach utilized two bi-temporal hyperspectral datasets captured by the Hyperion sensor, encompassing diverse land cover classes.

The results obtained from the proposed method were compared against reference data and two state-of-the-art hyperspectral change detection (HCD) algorithms. Notably, the proposed method demonstrated superior accuracy and a reduced false alarm (FA) rate. The average overall accuracy (OA) and Kappa coefficient (KC) exceeded 96% and 0.90, respectively, while the average FA rate remained below 5%.

This novel CD framework represents a breakthrough in spectral change detection, showcasing its effectiveness in discerning subtle alterations in land cover over time. The high accuracy and low FA rate underscore its potential as a reliable tool for monitoring and analyzing bi-temporal hyperspectral imagery, contributing to advancements in change detection methodologies within the realm of remote sensing and environmental monitoring.

2.5. Navigating Urban Fringe Transformation: A GeoSOS-FLUS Model for Simulating and Predicting Land Use Changes in Daxing District, Beijing, China

The dynamic nature of land use changes in urban fringe areas demands effective modeling and prediction tools to inform urban development regulation. This study, centered in Daxing District, Beijing—a representative urban fringe of the city—employs the GeoSOS-FLUS model (geographical simulation and optimization system–future land use simulation) and Markov chain model to simulate and predict land use changes. Two periods of land use data from 2008 and 2018 served as the foundation for predictions in 2028 and 2038. The study also conducts scenario simulations to evaluate potential future trajectories.

The results from future predictions under various scenarios reveal compelling insights: (1) in the natural development scenario, construction land and grassland exhibit gradual increases, while cultivated land, woodland, and water bodies exhibit gradual decreases; (2) the cultivated land protection scenario maintains the cultivated land area, curbing the expansion of construction land, and fostering a slow increase in woodland and water bodies after an initial decline in grassland; (3) under the ecological control scenario, cultivated land, grassland, woodland, and water bodies show slow increasing trends, with a limited conversion of cultivated land to construction land.

These findings underscore the efficacy of scenario-based simulations in influencing land use patterns. The study demonstrates that scenarios emphasizing cultivated land protection and ecological control can effectively limit the expansion of construction land. By offering valuable insights into potential future urban development trajectories, this research serves as a foundational tool for informed decision making in the regulation and control of urban development in Daxing District, Beijing.

2.6. Transforming Change Detection: CD-TransUNet, a Hybrid Transformer Model for Urban Building Change Detection in L-Band SAR Images

The detection of changes in urban buildings is a pivotal area within remote sensing research, crucial for urban planning, disaster assessments, and surface dynamic monitoring. SAR images, distinguished by abundant information and substantial data volume, offer unique advantages over traditional optical images. However, existing SAR-based change detection methods for buildings often struggle with small building detection and exhibit poor edge segmentation. To address these challenges, this paper introduces CD-TransUNet, a novel deep learning approach designed for precise building change detection.

CD-TransUNet is an end-to-end encoding–decoding hybrid transformer model that seamlessly integrates UNet and transformer architectures. To enhance feature extraction precision and reduce computational complexity, CD-TransUNet incorporates coordinate attention (CA), atrous spatial pyramid pooling (ASPP), and depthwise separable convolution (DSC). Additionally, by directing differential images to the input layer, CD-TransUNet prioritizes building changes on a large scale while disregarding changes in other land types.

The proposed method's efficacy is validated using a pair of ALOS-2(L-band) acquisitions, with comparative experimental results against baseline models demonstrating significantly higher precision and a Kappa value reaching 0.795. Noteworthy attributes, such as low missed alarms and accurate building edge detection, affirm the suitability of CD-TransUNet for building change detection tasks. This research presents a powerful tool for advancing change detection in urban environments, particularly when working with L-Band SAR images.

2.7. Guiding Sustainable Urban Expansion: Utilizing SVM-Based Simulation to Establish Urban Growth Boundaries in Chattogram, Bangladesh

In the face of rapid and unregulated urban expansion in Chattogram, Bangladesh, this study advocates for the implementation of urban growth restriction mechanisms, particularly the urban growth boundary (UGB), to channel development away from environmentally sensitive areas. Leveraging a support vector machine (SVM)-based urban growth simulation model, this paper delves into the identification of future contiguous expansion areas in the city, with the ultimate goal of delineating an effective UGB.

The SVM model, utilizing the radial basis function (RBF) kernel, is constructed using land cover, topographic, and population density data spanning the past two decades. Fourteen predictor variables are employed to develop a robust model. A grid-search is performed to fine-tune hyperparameters, optimizing the RBF kernel function's performance in the SVM. The finalized SVM model, with the best-performing hyperparameter combination, demonstrates an impressive 91.79% agreement and substantial agreement with a Kappa coefficient of 0.7699.

The SVM simulation model not only identifies areas prone to urban expansion in Chattogram over the next two decades, but also provides essential insights for the stringent delineation of a UGB. This research presents a valuable tool for urban planners and policymakers, enabling informed decisions to guide sustainable urban growth and protect environmentally sensitive regions in Chattogram, Bangladesh.

2.8. Navigating Urban Dynamics: Modeling Growth and Climate Change Impacts in Esmeraldas City, Ecuador

Situated in one of Ecuador's economically challenged urban centers, Esmeraldas faces a unique confluence of historical reliance on natural resource extraction and limited investment in local populations. This research aims to contribute valuable insights by creating a predictive scenario for urban growth in Esmeraldas, intricately linked to future climate projections and emphasizing vulnerability to landslides and flooding. Methodological advancements are sought through the integration of urban growth simulations and the downscaling of global climate change models.

Spatially explicit simulations, specifically cellular automata (CA), serve as the cornerstone for capturing the dynamic processes of urban growth. CA are intricately linked to vulnerability analysis based on socioeconomic conditions, with a specific focus on areas susceptible to flooding and landslides. The study uncovers a positive relationship between the proportion of Afro-Ecuadorian residents and the risk of landslides and flooding with urban growth. Projections based on future scenarios indicate a 50% increase in the urban growth area compared with 2016 if current trends persist. Moreover, the removal of natural vegetation, including mangroves, raises concerns about heightened vulnerability to climate change.

This research not only sheds light on the intricate dynamics of urban growth in Esmeraldas, but also serves as a methodological guide for integrating urban simulation and climate change considerations. By identifying critical links between socioeconomic conditions, vulnerability, and urban growth, the study provides a foundation for informed decision making in urban development and climate resilience planning in Esmeraldas, Ecuador.

2.9. Navigating Urban Evolution: Multisource Data Analysis for Spatial Development—A Case Study of Xianyang City's Integration into the Xi'an International Metropolis

The study of urban spatial development delves into the intricate process of urbanization, encompassing economic dynamics, population shifts, urban construction land scale, and the structure of construction land, all of which collectively shape the economic, social, and functional structures of a city. Focusing on Xianyang City, a pivotal component of the Xi'an international metropolis, this research leverages a comprehensive dataset, including night light remote sensing data from 1992 to 2013, land use data spanning six periods from 1980 to 2015, AutoNavi Map (AMAP) points of interest (POI) data, and the patch-generated land use simulation model (PLUS). The objective is to simulate the spatial-temporal pattern change characteristics of land use in Xianyang City from 2025 to 2035.

The key findings include (1) a significant upward trend in urban land use from 1985 to 2015; (2) the gravitational center of Xianyang City's built-up area shifting southeast and then northeast from 1992 to 2013, accelerating notably after 2010; (3) consistent distribution patterns of different urban centers in Xianyang City; (4) anticipated trends from 2005 to 2035, including multipolar explosive growth in construction land, slow growth in forest land, and an initial decrease followed by an increase in wetland water bodies. This indicates

a shift from a single-center development model to a point-axis development model in the urban spatial structure.

This study provides valuable insights into the evolving urban landscape, offering reference points for future urban construction layouts in Xianyang City. The multifaceted analysis, incorporating diverse data sources, contributes to a delicate understanding of the complex dynamics shaping urban spatial development in the context of Xi'an international metropolis integration.

2.10. Navigating Urbanization: Assessing the Evolution of Green Spaces Amid Rapid Urban Expansion in Southeast Asian Cities

The phenomenon of rapid urban expansion has led to a significant decline in green spaces within urban areas globally. Focusing on three Southeast Asian cities—Kuala Lumpur City, Malaysia; Jakarta, Indonesia; and Metro Manila, Philippines—this study explores the spatial and temporal patterns of urban areas and green space structures over the past two decades. Land use land cover (LULC) maps for the cities in 1988/1989, 1999, and 2014 were developed using 30 m resolution satellite images. Changes in landscape and spatial structure were analyzed through change detection, landscape metrics, and statistical analysis.

Over the 25-year period, the proportion of green space in these cities diminished from 45% to 20% due to rapid urban expansion. Notably, in Metro Manila and Jakarta, a higher proportion of green space was converted to urban areas during the initial 1989 to 1999 period than in the subsequent 1999 to 2014 period. Significant changes in green space structure were observed in Jakarta and Metro Manila, characterized by fragmentation, reduced connectivity, and uneven distribution. Conversely, Kuala Lumpur City did not exhibit such changes.

The study underscores the influence of spatial structure and population density on green space, with Jakarta and Metro Manila experiencing a higher impact compared with Kuala Lumpur. These findings contribute valuable insights into the relative contributions of green space structure in Southeast Asian cities undergoing rapid urbanization.

2.11. Modeling Delhi's Urban Landscape: Simulation of Urban Expansion Using a Logistic Regression Model Based on Various Driving Factors

Over the last three decades, Delhi has undergone extensive and rapid urban expansion, especially in the East and South East zone, resulting in a substantial increase in the total built-up area from 195.3 sq. km to 435.1 sq. km between 1989 and 2020. This expansion has led to habitat fragmentation, deforestation, and challenges in efficiently managing urban utility services in the newly extended areas. This research aims to simulate urban expansion in Delhi by employing a logistic regression model that considers various driving factors.

Recent urban expansion in Delhi was mapped using LANDSAT images from 1989, 2000, 2010, and 2020. The analysis employed concentric rings to illustrate the intensity of urban expansion in different directions. Nine driving factors were evaluated to discern their influence on the urban expansion process. Proximity to urban areas, main roads, and medical facilities emerged as the most significant factors during 1989–2020, with the highest regression coefficients being -0.884 , -0.475 , and -0.377 , respectively. The predicted pattern of urban expansion exhibited chaotic, scattered, and dense natures on the peripheries, potentially leading to further losses of natural resources.

To assess the accuracy of the simulation, the relative operating characteristic method was applied, yielding a value of 0.96, affirming the validity of the simulation. The research outcomes provide valuable insights for local authorities to recognize future expansion patterns, facilitating the implementation of effective policies for achieving sustainable urban development in Delhi.

2.12. Navigating Urban Futures: A Comprehensive Review of Land Use/Land Cover Change Modeling for Urban Development

Land use land cover (LULC) modeling represents a crucial tool for understanding the intricacies of future urban expansion. This paper offers a comprehensive review of existing LULC modeling techniques and novel approaches employed by the research community. The review undertakes a comparative analysis of each technique, exploring their applications, utility, drawbacks, and broader differences. The aim is to underscore the strengths and weaknesses of individual techniques, emphasizing the need for a delicate understanding.

Furthermore, the review highlights the potential of hybridizing different techniques, such as combining machine learning models with statistical models, to enhance the effectiveness of LULC modeling by leveraging their respective strengths. Despite significant progress in LULC modeling, the review emphasizes the importance of integrating a policy framework into the modeling process to enhance urban planning and management. This integration is crucial for achieving better land management practices and contributing to the realization of Sustainable Development Goal-15 (SDG-15), focused on life on land.

This comprehensive review serves as a valuable resource for researchers and policymakers alike, providing insights that can contribute to improved land management practices and, ultimately, aid in achieving sustainable urban development objectives.

3. Conclusions

As we reflect on the insights gleaned from this Special Issue, it becomes evident that urban expansion is a dynamic and multifaceted phenomenon that necessitates a continued approach to understanding its dynamic. The articles presented here serve as beacons of knowledge, collectively illuminating the path toward sustainable urban development. Nevertheless, the journey is far from over.

The integration of AI into the realm of urban expansion research offers an exciting frontier. It is not merely about adopting advanced technologies, but about reimagining how we perceive, analyze, and plan for urban growth. AI has the potential to transcend existing boundaries, offering a more holistic understanding of urban dynamics and, consequently, empowering us to build cities that are not only expansive but sustainable, resilient, and adaptive.

As we embark on this journey into the future, let us embrace the transformative power of AI in shaping our urban landscapes. The challenges are immense, but so are the possibilities. By uniting the precision of AI with the wisdom gained from traditional and contemporary research, we can lay the foundation for a new era of urban development—one that is not just expansive but also enlightened, equitable, and enduring. The future of urban expansion is in our hands, guided by the synergy of human intellect and artificial intelligence.

Conflicts of Interest: The authors declare no conflict of interest.

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